

Summary of Backpropagation-Free Training

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1 Memory Consumption

Before talking about split learning, I would like to supplement last week’s conclusion with additional information regarding memory consumption. In the original implementation by [1], the activation values for each model layer were not saved. This approach significantly reduced memory usage compared to scenarios where the activation values are preserved. Table 1 records the memory consumption under different conditions.

Table 1: Memory usage for different methods.

Training Strategy	Model	Memory Usage (MB)
Back-Propagation	ResNet 5	584
	VGG 8	596
ULR	ResNet 5 (With Activation)	20076
	ResNet 5 (Without Activation)	35908
	ResNet 5 (Extreme Case)	572
	VGG 8 (With Activation)	15638
	VGG 8 (Without Activation)	29881
	VGG 8 (Extreme Case)	590

2 Split Learning with ULR

I successfully implemented the ULR method for split learning. However, regardless of whether the data is independently and identically distributed (IID) or non-IID, the accuracy of the ULR method is generally lower than that of the backpropagation-based approach in most cases.

The dataset used in the experiment is CIFAR-10. For the non-IID scenario, the data was allocated among clients according to a specific ratio α , ensuring that each client received an equal amount of data. Table 2 provides a detailed comparison of the accuracy achieved by both methods under various conditions.

3 Reflection and planning

As observed in previous experiments, split learning tends to yield lower accuracy, particularly when dealing with non-IID data. To address this limitation, I plan to incorporate model aggregation into the ULR approach. This modification aims to create a more flexible training framework while also improving overall accuracy.

Table 2: Accuracy of split learning for different methods.

Non-IID Setting	Model	Accuracy(%)
$\alpha = 0$	ResNet 5 (ULR)	52.18
	ResNet 5 (BP)	70.94
	VGG 8 (ULR)	64.82
	VGG 8 (BP)	76.84
$\alpha = 0.5$	ResNet 5 (ULR)	49.55
	ResNet 5 (BP)	67.22
	VGG 8 (ULR)	59.50
	VGG 8 (BP)	71.58
$\alpha = 0.95$	ResNet 5 (ULR)	23.17
	ResNet 5 (BP)	27.93
	VGG 8 (ULR)	14.24
	VGG 8 (BP)	10.00

References

- [1] J. Jiang, Z. Zhang, C. Xu, Z. Yu, and Y. Peng, “One forward is enough for neural network training via likelihood ratio method,” *arXiv preprint arXiv:2305.08960*, 2023.